

Final formulation

Sets and parameters:

D	ordered set of all days of the scheduling period, where $d = 1$ is the first day and $d = D $ is the last day of the scheduling period
$D_o \subset D$	set of shop open days
$\bar{D} \subset D$	set of days $d \in D$ such that all days in the interval from d up to $d + 5$ (inclusive) belong to D_o
T	ordered set of all possible timeslots in any day, where $t = 1$ is the first timeslot of the day and $t = T $ is the last time slot of the day
$T_d \subset T$	set of timeslots that must be covered in day $d \in D$
W	set of all workers $w \in W$
K	ordered set of weeks of the scheduling period, where $k = 1$ is the first week and $k = K $ is the last week of the scheduling period
$D_k \subset D_o$	set of open days of week $k \in K$
$D_{wk} \subset D_k$	set of preferable variable days-off of worker $w \in W$ in week $k \in K$ (this set might be empty for some workers on some weeks)
$U_{wk} \subset D_k$	set of unavailable days of worker $w \in W$ in week $k \in K$, i.e., days that worker cannot work in week k due to holydays, fixed days-off or other unavailability reason
n_{wk}	number of working days that must be assigned to worker $w \in W$ in week $k \in K$ (equal to the number of open days in week k minus the number of unavailable days of worker w in week k minus the number of preferred days-off of worker w in week k)
S	set of all skills $s \in S$
L	ordered set of all competence levels among all skills, where $l = 1$ represents the highest level and $l = L $ represents the lowest level
$S_w \subset S$	set of skills of worker $w \in W$; for each $s \in S_w$, parameter $l_{ws} \in L$ indicates the competence level of worker $w \in W$ in skill $s \in S_w$
H_{wd}	set of all possible daily working assignments for worker $w \in W$ in day $d \in D$; each assignment $h \in H_{wd}$ is defined by the binary parameters δ_{wdht} indicating if timeslot $t \in T_d$ is covered by the daily working assignment $h \in H_{wd}$ for worker $w \in W$ in day $d \in D$ (H_{wd} does not include assignments with timeslots t not belonging to T_d)
α_{dts}	minimum number of workers desired on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$
p_{sl}	priority weight of skill $s \in S$ and competence level $l \in L$ (for example, if $(s, l) = (\text{Storage}, 1)$ has higher priority than $(\text{Chackout}, 2)$, then $p_{\text{Storage},1} \leq p_{\text{Storage},2}$)

Variables:

x_{wdh}	binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in daily working assignments $h \in H_{wd}$
x'_{wd}	binary variable that, when equal to 1, indicates that worker $w \in W$ works on a day $d \in D_{wk}$ (i.e., a preferable day-off of worker $w \in W$)
y_{wdts}	binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$

y'_{wdtsl}	binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$ and competence level $l \in L$
z_{dts}	integer variable with the number of workers below the desired value α_{dts} on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$

Objective functions:

ObjectiveFunction1: aiming to fulfil as much as possible the minimum desired number α_{dts} of workers in all days, timeslots and skills

$$\text{Minimize } \sum_{d \in D_o} \sum_{t \in T_d} \sum_{s \in S} z_{dts} \quad (1)$$

The aim is to minimize the sum of the number of workers below the desired value α_{dts} in all open days $d \in D_o$, timeslots $t \in T_d$ and skills $s \in S$.

ObjectiveFunction2: aiming to assign preferably multi-skilled workers to higher-priority skill-level combinations

$$\text{Minimize } \sum_{w \in W} \sum_{d \in D_o} \sum_{t \in T_d} \sum_{s \in S_w} \sum_{l \in L} p_{sl} \times y'_{wdtsl} \quad (1')$$

The aim is to minimize the total priority weight assigned to all workers $w \in W$ in all open days $d \in D_o$, all timeslots $t \in T_d$ and all skills $s \in S_w$.

ObjectiveFunction3: aiming to assign as much as possible the preferable days-off of all workers

$$\text{Minimize } \sum_{w \in W} \sum_{d \in D_o} x'_{wd} \quad (1'')$$

The aim is to minimize the number of preferable days-off of all workers that are assigned as working days.

Constraints:

$$\sum_{h \in H_{wd}} x_{wdh} \leq 1 \quad , w \in W, d \in D_o \quad (2)$$

With constraints (2), each worker $w \in W$ can be assigned on each open day $d \in D_o$ with at most one working daily assignment $h \in H_{wd}$.

$$\sum_{s \in S_w} y_{wdts} = \sum_{h \in H_{wd}} \delta_{wdht} x_{wdh} \quad , w \in W, d \in D_o, t \in T_d \quad (3)$$

With constraints (3), each worker $w \in W$ assigned on open day $d \in D_o$ with a working daily assignment $h \in H_{wd}$ that covers timeslot $t \in T_d$ (i.e., a timeslot such that parameter $\delta_{wdht} = 1$) must be assigned in timeslot t with one of its skills S_w .

$$\sum_{d=d'}^{d'+5} \sum_{h \in H_{wd}} x_{wdh} \leq 5 \quad , w \in W, d' \in \bar{D} \quad (4)$$

With constraints (4), each worker $w \in W$ cannot be assigned with more than 5 working days in any set of 6 consecutive open days.

$$\sum_{h \in H_{wd}} x_{wdh} = 0 \quad , w \in W, k \in K, d \in U_{wk} \quad (5)$$

With constraints (5), each worker $w \in W$ cannot be assigned with any working daily assignment in the days $d \in U_{wk}$ (i.e., the days he is unavailable) on each week $k \in K$.

$$\sum_{d \in D_k} \sum_{h \in H_{wd}} x_{wdh} = n_{wk} \quad , w \in W, k \in K \quad (6)$$

With constraints (6), each worker $w \in W$ must be assigned on each week $k \in K$ with n_{wk} working days.

NOTE: The previous constraints from (2) to (6) guarantee a feasible solution (i.e., they guarantee the hard timetable constraints). The next constraints form (7) to (9) are required to define the objective functions.

$$z_{dts} + \sum_{w \in W} y_{wdts} \geq \alpha_{dts} \quad , d \in D_o, t \in T_d, s \in S \quad (7)$$

With constraints (7), the value of variable z_{dts} is at least the number of workers below the required value α_{dts} for each open day $d \in D_o$, each timeslot $t \in T_d$ and each skill $s \in S$. Variables z_{dts} are used in **ObjectiveFunction1** aiming to fulfil as much as possible the minimum desired number of workers in all days, timeslots and skills.

$$y'_{wdtsl} = \begin{cases} y_{wdts} & , l = l_{ws} \\ 0 & , l \neq l_{ws} \end{cases} \quad , w \in W, d \in D_o, t \in T_d, s \in S_w \quad (8)$$

With constraints (8), each worker $w \in W$ that works in open day $d \in D_o$ in time slot $t \in T$ with skill $s \in S$ is assigned with its competence level l_{ws} and the variables associated to all other competence levels are set to zero. Variables y'_{wdtsl} are used in **ObjectiveFunction2** aiming to assign preferably multi-skilled workers to higher-priority skill-level combinations.

$$x'_{wd} = \begin{cases} \sum_{h \in H_{wd}} x_{wdh} & , d \in D_{wk} \\ 0 & , d \notin D_{wk} \end{cases} \quad , w \in W, k \in K, d \in D_k \quad (9)$$

With constraints (9), variable x'_{wd} is set to 1 when worker $w \in W$ works on a day $d \in D_{wk}$ (i.e., a preferable day-off of worker $w \in W$) and is set to 0 in all other cases. Variables x'_{wd} are used in **ObjectiveFunction3** aiming to assign as much as possible the preferable days-off of all workers.